

SAI SATHYA ANDRAJU

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PROFESSIONAL SUMMARY

Results-driven **Machine Learning Engineer** with experience of designing and deploying **AI/ML and LLM-based systems** at scale. Proven success in optimizing **large language models**, reducing inference latency by 30%, and improving GPU utilization by 25%. Skilled in **MLOps, distributed computing, data engineering, and RAG architectures** using **Python, TensorFlow, PyTorch, and AWS**. Experienced in delivering enterprise-grade ML systems for **Fintech and Healthcare** clients with measurable accuracy and efficiency gains. Certified in **Databricks ML, AWS ML, and TensorFlow Developer**, passionate about production-grade AI solutions.

CORE COMPETENCIES

Machine Learning • Deep Learning • LLM Optimization • Generative AI • RAG • TensorFlow • PyTorch • MLflow • Ray • Databricks • Spark • MLOps • AWS SageMaker • Docker • Kubernetes • FastAPI • NLP • API Development • Data Pipelines • Model Deployment • Feature Engineering • CI/CD • GPU Optimization • Cloud Infrastructure • Agile/Scrum

Technical Skills

Programming Languages: Python, SQL, JavaScript, React.js

Machine Learning & AI: TensorFlow, PyTorch, Scikit-learn, XGBoost, LightGBM, MLflow, Ray, Hugging Face Transformers, LangChain, LLM Fine-tuning, RAG

Data Engineering & Processing: Apache Spark, Delta Lake, Delta Live Tables, Apache Airflow, ETL/ELT Pipelines, Feature Engineering, Data Validation

MLOps & Deployment: Docker, Kubernetes, AWS (EC2, Lambda, S3, SageMaker), FastAPI, Flask, CI/CD, Model Versioning, A/B Testing, Model Monitoring

Databases & Storage: PostgreSQL, MongoDB, Redis, Vector Databases, Databricks Feature Store, Unity Catalog

Big Data & Analytics: Databricks, PySpark, Pandas, NumPy, Data Quality Frameworks, Schema Evolution

Natural Language Processing: spaCy, NLTK, Tesseract OCR, OpenCV, Tokenization, Named Entity Recognition (NER), Text Classification

Tools & Platforms: Git, Jupyter Notebooks, VS Code, Postman, Pytest, JIRA, Agile/Scrum

AI Governance & Compliance: Model Lineage Tracking, Data Privacy, Regulatory Compliance (GDPR, HIPAA), Model Interpretability (SHAP, LIME)

Optimization Techniques: Kernel Fusion, Model Quantization, Distributed Training, GPU Optimization, Caching Strategies

PROFESSIONAL EXPERIENCES

Software Engineer

Databricks – San Francisco, CA | Dec 2024 – Present

- Designed and deployed **large-language-model (LLM) fine-tuning pipelines** using Databricks Mosaic AI, enabling secure customization for enterprise clients.
- Optimized **model serving performance** with Ray, MLflow, and Delta Lake, reducing latency by **30%** and improving GPU utilization by **25%**.
- Enhanced MLflow Tracking Server with **asynchronous logging and version control**, accelerating experiment tracking by **25%**.
- Built **real-time feature pipelines** using Delta Live Tables for fraud detection and recommendations, cutting pipeline failures by **40%**.

- Partnered with AI Governance to extend **Unity Catalog** for model lineage and compliance metadata, improving audit readiness in regulated sectors.
- Contributed to **DBRX optimization**, implementing kernel fusion and quantization that doubled inference speed and reduced memory by **30%**.
- Automated **feature lineage and version tracking** within Databricks Feature Store, increasing feature reuse by **35%**.
- Collaborated across product, research, and data teams to develop **RAG (Retrieval-Augmented Generation)** pipelines using Vector Search, enhancing accuracy for enterprise AI solutions.

Software Developer

EDB – Pune, India | Aug 2021 – Aug 2022

- Designed and deployed **AI/ML models** for Fintech and Healthcare clients using Python, TensorFlow, and Scikit-learn, turning analytics into production solutions.
- Built an **AI-powered Credit Risk Scoring System**, automating loan approvals and improving accuracy by **18%** while reducing review time by **25%**.
- Developed a **Disease Prediction Platform** using XGBoost and NLP-based extraction, enabling early diagnosis and preventive-care analytics.
- Engineered **end-to-end ML pipelines** covering preprocessing, feature engineering, training, validation, and deployment via FastAPI and AWS Lambda.
- Implemented **MLOps workflows** with Docker, Git, and AWS S3 to automate retraining, versioning, and CI/CD.
- Partnered with cross-functional teams to integrate predictive APIs into client web and mobile applications.
- Conducted **A/B testing**, interpretability analysis (SHAP, LIME), and drift monitoring to ensure model reliability and regulatory compliance.
- Authored technical documentation and research presentations to communicate AI outcomes to business stakeholders.

Software Engineering Intern

Gartner – Chennai, India | Jan 2021 – July 2021

- Contributed to ML projects including **Churn Prediction, Invoice Automation, and Recommendation Systems** using Python, Scikit-learn, and TensorFlow.
- Built **OCR/NLP-based invoice processing system** with Tesseract, OpenCV, and spaCy achieving **94% extraction accuracy**.
- Developed and integrated **REST APIs (Flask/FastAPI)** and React.js components for deploying ML models into applications.
- Automated **ETL data pipelines** with PostgreSQL and MongoDB for ingestion and feature engineering.
- Deployed ML models on **AWS EC2** with Docker and Redis caching for faster performance.
- Participated in Agile sprints, conducted unit testing with Pytest, and presented technical demos to stakeholders.

EDUCATION

Master of Science in Information Technology Management

St. Francis College, Brooklyn, NY | Jan 2024 – May 2025

Bachelor of Science in Computer Science

Lovely Professional University, Phagwara, India | Aug 2018 – May 2022

CERTIFICATIONS

- Databricks Certified Machine Learning Professional
- AWS Certified Machine Learning – Specialty
- TensorFlow Developer Certificate